

Práctica 2

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1. Introducción

- **Objetivo de la práctica:** Familiarizarse con herramientas de monitorización en Unix (Ubuntu) y analizar el rendimiento del sistema bajo distintas condiciones.
- Breve descripción del contexto: Máquina usada (física o virtual), distribución Ubuntu, especificaciones básicas del sistema.

2. Descripción General del Sistema Bajo Estudio

Especificaciones técnicas del fabricante

Category	Specification
Product Number	RZ09-0421E*D3, RZ09-0421N*C3, RZ09-0421N*D3, RZ09-0421P*C3, RZ09-0421P*D3, RZ09-0421P*F3, RZ09-0421N*G3
Operating System	Windows 11 Home
Processor	12th Gen Intel Core i7 Processor (14-core) i7-12800H, 1.8 GHz with Turbo Boost up to 4.8 GHz, with 24 MB of Cache
Graphics	NVIDIA® GeForce RTX™ 3070 Ti Laptop GPU
GPU Features	• NVIDIA® GeForce RTX™ 3070 Ti Laptop GPU • 8 GB GDDR6 VRAM • NVIDIA CUDA® Cores - 5888 • Maximum Graphics Power up to 100 W • Boost Clock up to 1200 MHz • NVIDIA GPU Boost™ 2.0

	• NVIDIA Optimus™ Technology • NVIDIA Whisper Mode 2.0 • NVIDIAResizable BAR • 2nd Gen Ray Tracing Cores • 3rd Gen Tensor Cores • VR Ready
Display	15.6" IPS QHD 240 Hz, 2560 × 1440 native resolution with a 2.5 ms response rate support, Advanced Optimus and NVIDIA® G-SYNC®, individually factory calibrated
Thermal System	Vapor Chamber Cooling
Storage	• 1 TB PCIe Gen 4 NVMe upgradeable to 2 TB SSD • Open M.2 Slot - Supports PCIe SSDs, upgradeable to 2 TB SSD (Dual M.2 Slot – Supports Single Sided M.2 Only)
Memory	16 GB DDR5 4800 MHz Dual-Channel Memory Upgradeable to 64 GB
Input & Output	• Precision glass touchpad • Per key RGB Powered by Razer Chroma™ N-Key rollover keyboard • 3 x USB 3.2 Gen 2 Type-A Ports • 1 x USB 3.2 Gen 2 Type-C Port and 1 x Thunderbolt™ 4 Port with Power Delivery and Display Port 1.4 • Charging supported with 20 V USB-C chargers with PD 3.0" • HDMI 2.1 output
Communication	• Wi-Fi 6E, IEEE 802.11a/b/g/n/ac/ax/az • Bluetooth® 5.2
Audio	• Built-in stereo speakers • 3.5 mm headphone/microphone combo port • Built-in array microphone • THX® Spatial Audio • 7.1 Codec support (via HDMI)
Additional Features	• Razer™ Synapse 3 enabled with performance, programmable keyboard, backlighting, and fan control • Kensington™ Security Slot • Modern Standby • Intel PTT - (sTPM 2.0)
Battery & Adaptor	• Built-in 80 WHr lithium polymer – Limited 2 Year Warranty • Compact 230 W Power Adapter

Keyboard & Trackpad	• Per-Key RGB powered by Razer Chroma™ anti-ghosting backlit keyboard • Glass Touchpad with Microsoft Precision Touch for multi-touch gestures
Camera	Windows Hello IR Full HD webcam 1080P (F/2 Lens)
Chassis	CNC Aluminum, Anodized Finish with Fingerprint Resistive Coating
Color	Black with illuminated Razer logo
Approximate Dimensions	• H: 0.67 in / 16.99 mm • W: 13.98 in / 355 mm • D: 9.25 in / 235 mm
Approximate Weight	2.02 kg / 4.4 lbs

Especificaciones técnicas (desde terminal):

Procesador (`uname -a` , `/proc/cpuinfo`): Muestra información general del sistema operativo, versión del Kernel, arquitectura y nombre del equipo.

Proporciona información detallada sobre la arquitectura del procesador (modelo, núcleos, frecuencia, caché, etc.).

```
lucian@lucian-Blade-15-2022-RZ09-0421:~$ uname -a
Linux lucian-Blade-15-2022-RZ09-0421 6.11.0-18-generic #18-Ubuntu SMP
```

```
lucian@lucian-Blade-15-2022-RZ09-0421:~$ cat /proc/cpuinfo
processor       : 0
vendor_id     : GenuineIntel
cpu family    : 6
model        : 154
model name    : 12th Gen Intel(R) Core(TM) i7-12800H
stepping      : 3
microcode    : 0x436
cpu MHz      : 400.000
cache size   : 24576 KB
physical id  : 0
siblings     : 20
core id      : 0
```

```

cpu cores   : 14
apicid      : 0
initial apicid : 0
fpu         : yes
fpu_exception : yes
cpuid level : 32
wp          : yes
flags       : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cm
vmx flags   : vnmi preemption_timer posted_intr invvpid ept_x_only ept_ad
bugs        : spectre_v1 spectre_v2 spec_store_bypass swapgs eibrs_pbrsb r
bogomips    : 5606.40
clflush size : 64
cache_alignment : 64
address sizes : 46 bits physical, 48 bits virtual
power management:

```

Memoria RAM disponible (`free -h` , `/proc/meminfo`): Muestra la cantidad total, usada y disponible de memoria RAM y Swap en formato legible por humanos.
Muestra la cantidad total, usada y disponible de memoria RAM y Swap en formato legible por humanos.

```

lucian@lucian-Blade-15-2022-RZ09-0421:~$ free -h
              total        usado        libre compartido  búf/caché disponible
Mem:          14Gi       4,8Gi       4,5Gi       802Mi       6,6Gi       9Gi
Inter:         4,0Gi         0B       4,0Gi

```

```

lucian@lucian-Blade-15-2022-RZ09-0421:~$ cat /proc/meminfo
MemTotal:      15524496 kB
MemFree:       4289564 kB
MemAvailable: 10060608 kB
Buffers:       154664 kB
Cached:        6844764 kB
SwapCached:          0 kB
Active:        4316660 kB
Inactive:      4511308 kB
Active(anon):  3013848 kB

```

Inactive(anon): 0 kB
Active(file): 1302812 kB
Inactive(file): 4511308 kB
Unevictable: 992108 kB
Mlocked: 32 kB
SwapTotal: 4194300 kB
SwapFree: 4194300 kB
Zswap: 0 kB
Zswapped: 0 kB
Dirty: 68 kB
Writeback: 0 kB
AnonPages: 2820572 kB
Mapped: 781860 kB
Shmem: 1185284 kB
KReclaimable: 301660 kB
Slab: 665860 kB
SReclaimable: 301660 kB
SUnreclaim: 364200 kB
KernelStack: 20992 kB
PageTables: 48456 kB
SecPageTables: 7396 kB
NFS_Unstable: 0 kB
Bounce: 0 kB
WritebackTmp: 0 kB
CommitLimit: 11956548 kB
Committed_AS: 20195144 kB
VmallocTotal: 34359738367 kB
VmallocUsed: 156208 kB
VmallocChunk: 0 kB
Percpu: 20400 kB
HardwareCorrupted: 0 kB
AnonHugePages: 0 kB
ShmemHugePages: 942080 kB
ShmemPmdMapped: 0 kB
FileHugePages: 0 kB
FilePmdMapped: 0 kB

```

Unaccepted:      0 kB
HugePages_Total: 0
HugePages_Free:  0
HugePages_Rsvd:  0
HugePages_Surp:  0
Hugepagesize:    2048 kB
Hugetlb:         0 kB
DirectMap4k:     814884 kB
DirectMap2M:     7249920 kB
DirectMap1G:     8388608 kB
lucian@lucian-Blade-15-2022-RZ09-0421:~$

```

Almacenamiento (`df -h`): Información más detallada sobre el uso de memoria RAM del sistema operativo, incluyendo buffers, cachés, memoria libre y asignada.

```

lucian@lucian-Blade-15-2022-RZ09-0421:~$ df -h
S.ficheros  Tamaño Usados  Disp Uso% Montado en
tmpfs       1,5G  2,9M  1,5G   1% /run
/dev/nvme0n1p6 170G  68G  94G  42% /
tmpfs       7,5G  76M  7,4G   1% /dev/shm
efivarfs    192K  125K  63K  67% /sys/firmware/efi/efivars
tmpfs       5,0M  12K  5,0M   1% /run/lock
tmpfs       1,0M   0  1,0M   0% /run/credentials/systemd-journald.service
tmpfs       1,0M   0  1,0M   0% /run/credentials/systemd-udev-load-cre
tmpfs       1,0M   0  1,0M   0% /run/credentials/systemd-tmpfiles-setup
tmpfs       1,0M   0  1,0M   0% /run/credentials/systemd-tmpfiles-setup
tmpfs       1,0M   0  1,0M   0% /run/credentials/systemd-sysctl.service
/dev/nvme0n1p2  96M  52M  45M  54% /boot/efi
tmpfs       7,5G  32M  7,4G   1% /tmp
tmpfs       1,0M   0  1,0M   0% /run/credentials/systemd-tmpfiles-setup
tmpfs       1,0M   0  1,0M   0% /run/credentials/systemd-resolved.servi
tmpfs       1,5G  144K  1,5G   1% /run/user/1000

```

Versión del kernel y distribución (`uname -r`):

```
lucian@lucian-Blade-15-2022-RZ09-0421:~$ uname -r
6.11.0-18-generic
```

3. Metodología y Condiciones del Experimento

Condiciones iniciales:

Estado inicial (ocioso): Sin programas adicionales abiertos o mínima carga (60s).

Estado bajo carga: Hemos usado el ejemplo propuesto con `ffmpeg` para transcodificar un vídeo durante el tiempo que ha necesitado el programa (~270s).

Los usuarios conectados han sido sólomente 2 durante todo el proceso de pruebas.

```
lucian@lucian-Blade-15-2022-RZ09-0421:~$ who
lucian  seat0      2025-03-18 15:44 (login screen)
lucian  tty2       2025-03-18 15:44 (tty2)
```

3.1. Preparativos

Primero hemos optado por instalar el paquete recomendado durante la práctica.

```
lucian@lucian-Blade-15-2022-RZ09-0421:~$ sudo apt-get install sysstat
```

Una vez instalado y configurado hemos procedido a probar los diferentes estados.

3.2 Estado inicial

```
lucian@lucian-Blade-15-2022-RZ09-0421:~$ sar -u -r -S 1 60
Linux 6.11.0-18-generic (lucian-Blade-15-2022-RZ09-0421)  18/03/25  _x86_64

16:55:59      CPU   %user   %nice   %system   %iowait   %steal   %idle
16:56:00    all    0,76    0,00    0,50    0,00    0,00   98,74
```

```

16:55:59  kbmemfree  kbavail kbmemused  %memused kbbuffers  kbcached k
16:56:00    2450304  8868196  4560912   29,38   227140  7566380  239270

16:55:59  kbswpfree kbswpused  %swpused  kbswpcad  %swpcad
16:56:00    4194300      0    0,00      0    0,00

16:56:00    CPU   %user   %nice  %system  %iowait  %steal   %idle
16:56:01    all   1,06   0,00   0,40   0,15   0,00   98,39

16:56:00  kbmemfree  kbavail kbmemused  %memused kbbuffers  kbcached k
16:56:01    2448284  8866172  4562908   29,39   227164  7566380  239270

16:56:00  kbswpfree kbswpused  %swpused  kbswpcad  %swpcad
16:56:01    4194300      0    0,00      0    0,00

16:56:01    CPU   %user   %nice  %system  %iowait  %steal   %idle
16:56:02    all   1,16   0,00   0,35   0,00   0,00   98,49

16:56:01  kbmemfree  kbavail kbmemused  %memused kbbuffers  kbcached k
16:56:02    2461692  8879580  4560788   29,38   227164  7555092  239158

16:56:01  kbswpfree kbswpused  %swpused  kbswpcad  %swpcad
16:56:02    4194300      0    0,00      0    0,00

16:56:02    CPU   %user   %nice  %system  %iowait  %steal   %idle
16:56:03    all   1,25   0,00   0,60   0,00   0,00   98,14

16:56:02  kbmemfree  kbavail kbmemused  %memused kbbuffers  kbcached k
16:56:03    2477748  8895636  4559176   29,37   227164  7540648  239013

16:56:02  kbswpfree kbswpused  %swpused  kbswpcad  %swpcad
16:56:03    4194300      0    0,00      0    0,00

16:56:03    CPU   %user   %nice  %system  %iowait  %steal   %idle
16:56:04    all   0,90   0,00   0,80   0,00   0,00   98,30

```

```

16:56:03 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:04 2479824 8897712 4558160 29,36 227164 7539588 239002

16:56:03 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:04 4194300 0 0,00 0 0,00

16:56:04 CPU %user %nice %system %iowait %steal %idle
16:56:05 all 0,85 0,00 0,50 0,00 0,00 98,64

16:56:04 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:05 2480336 8898224 4557648 29,36 227164 7539588 239002

16:56:04 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:05 4194300 0 0,00 0 0,00

16:56:05 CPU %user %nice %system %iowait %steal %idle
16:56:06 all 1,01 0,00 0,70 0,20 0,00 98,09

16:56:05 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:06 2487796 8905696 4569364 29,43 227172 7520404 238931

16:56:05 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:06 4194300 0 0,00 0 0,00

16:56:06 CPU %user %nice %system %iowait %steal %idle
16:56:07 all 1,06 0,00 0,40 0,00 0,00 98,54

16:56:06 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:07 2486840 8904744 4571680 29,45 227172 7519044 238924

16:56:06 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:07 4194300 0 0,00 0 0,00

16:56:07 CPU %user %nice %system %iowait %steal %idle
16:56:08 all 0,90 0,00 0,80 0,00 0,00 98,30

```

```

16:56:07 kbmemfree kbavail kbmemused %memused kbbuffers kbcached k
16:56:08 2484192 8902096 4573640 29,46 227172 7519732 2389314

16:56:07 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:08 4194300 0 0,00 0 0,00

16:56:08 CPU %user %nice %system %iowait %steal %idle
16:56:09 all 1,06 0,00 0,45 0,15 0,00 98,34

16:56:08 kbmemfree kbavail kbmemused %memused kbbuffers kbcached k
16:56:09 2491656 8909608 4567360 29,42 227188 7518532 2389194

16:56:08 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:09 4194300 0 0,00 0 0,00

16:56:09 CPU %user %nice %system %iowait %steal %idle
16:56:10 all 0,91 0,00 0,40 0,10 0,00 98,59

16:56:09 kbmemfree kbavail kbmemused %memused kbbuffers kbcached k
16:56:10 2492756 8910708 4566252 29,41 227196 7518532 2389194

16:56:09 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:10 4194300 0 0,00 0 0,00

16:56:10 CPU %user %nice %system %iowait %steal %idle
16:56:11 all 1,05 0,00 0,60 0,05 0,00 98,29

16:56:10 kbmemfree kbavail kbmemused %memused kbbuffers kbcached k
16:56:11 2490940 8908900 4566740 29,42 227196 7519860 238934

16:56:10 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:11 4194300 0 0,00 0 0,00

16:56:11 CPU %user %nice %system %iowait %steal %idle
16:56:12 all 0,95 0,00 0,65 0,00 0,00 98,40

```

```

16:56:11 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:12 2489916 8907876 4569092 29,43 227196 7518532 2389207

16:56:11 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:12 4194300 0 0,00 0 0,00

16:56:12 CPU %user %nice %system %iowait %steal %idle
16:56:13 all 1,01 0,00 0,35 0,00 0,00 98,64

16:56:12 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:13 2493112 8911072 4568968 29,43 227196 7515460 2388900.

16:56:12 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:13 4194300 0 0,00 0 0,00

16:56:13 CPU %user %nice %system %iowait %steal %idle
16:56:14 all 1,05 0,00 0,80 0,05 0,00 98,09

16:56:13 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:14 2486928 8904872 4570588 29,44 227204 7520016 238935

16:56:13 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:14 4194300 0 0,00 0 0,00

16:56:14 CPU %user %nice %system %iowait %steal %idle
16:56:15 all 0,70 0,00 0,50 0,00 0,00 98,79

16:56:14 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:15 2488184 8906140 4570820 29,44 227204 7518528 238920

16:56:14 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:15 4194300 0 0,00 0 0,00

16:56:15 CPU %user %nice %system %iowait %steal %idle
16:56:16 all 1,01 0,00 0,50 0,05 0,00 98,44

```

```

16:56:15 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:16 2490296 8908252 4568708 29,43 227204 7518528 238920

16:56:15 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:16 4194300 0 0,00 0 0,00

16:56:16 CPU %user %nice %system %iowait %steal %idle
16:56:17 all 1,01 0,00 0,55 0,00 0,00 98,44

16:56:16 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:17 2489384 8907340 4568356 29,43 227204 7519792 238933

16:56:16 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:17 4194300 0 0,00 0 0,00

16:56:17 CPU %user %nice %system %iowait %steal %idle
16:56:18 all 0,95 0,00 0,50 0,00 0,00 98,54

16:56:17 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:18 2490904 8908868 4568036 29,42 227204 7518592 238921

16:56:17 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:18 4194300 0 0,00 0 0,00

16:56:18 CPU %user %nice %system %iowait %steal %idle
16:56:19 all 1,20 0,00 0,70 0,10 0,00 97,99

16:56:18 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:19 2481036 8899000 4572484 29,45 227212 7523972 238956

16:56:18 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:19 4194300 0 0,00 0 0,00

16:56:19 CPU %user %nice %system %iowait %steal %idle
16:56:20 all 1,31 0,00 0,75 0,00 0,00 97,94

```

```

16:56:19 kbmempfree kbavail kbmempused %mempused kbbuffers kbcached k
16:56:20 2484572 8902536 4572944 29,46 227212 7519976 238916

16:56:19 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:20 4194300 0 0,00 0 0,00

16:56:20 CPU %user %nice %system %iowait %steal %idle
16:56:21 all 2,28 0,00 1,01 0,00 0,00 96,71

16:56:20 kbmempfree kbavail kbmempused %mempused kbbuffers kbcached k
16:56:21 2407900 8825868 4580712 29,51 227212 7588876 2396100

16:56:20 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:21 4194300 0 0,00 0 0,00

16:56:21 CPU %user %nice %system %iowait %steal %idle
16:56:22 all 1,31 0,00 0,66 0,00 0,00 98,03

16:56:21 kbmempfree kbavail kbmempused %mempused kbbuffers kbcached k
16:56:22 2427620 8845592 4573796 29,46 227212 7576072 239482

16:56:21 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:22 4194300 0 0,00 0 0,00

16:56:22 CPU %user %nice %system %iowait %steal %idle
16:56:23 all 1,21 0,00 0,40 0,10 0,00 98,29

16:56:22 kbmempfree kbavail kbmempused %mempused kbbuffers kbcached k
16:56:23 2416684 8834656 4571180 29,44 227228 7589608 239618

16:56:22 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:23 4194300 0 0,00 0 0,00

16:56:23 CPU %user %nice %system %iowait %steal %idle
16:56:24 all 0,76 0,00 0,60 0,05 0,00 98,59

```

```

16:56:23 kbmemfree kbavail kbmemused %memused kbbuffers kbcached k
16:56:24 2426848 8844836 4560916 29,38 227228 7589708 239619

16:56:23 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:24 4194300 0 0,00 0 0,00

16:56:24 CPU %user %nice %system %iowait %steal %idle
16:56:25 all 1,31 0,00 0,55 0,00 0,00 98,14

16:56:24 kbmemfree kbavail kbmemused %memused kbbuffers kbcached k
16:56:25 2438952 8856940 4550272 29,31 227228 7588248 239495

16:56:24 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:25 4194300 0 0,00 0 0,00

16:56:25 CPU %user %nice %system %iowait %steal %idle
16:56:26 all 0,81 0,00 0,45 0,00 0,00 98,74

16:56:25 kbmemfree kbavail kbmemused %memused kbbuffers kbcached k
16:56:26 2430148 8848136 4556308 29,35 227228 7591016 239523

16:56:25 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:26 4194300 0 0,00 0 0,00

16:56:26 CPU %user %nice %system %iowait %steal %idle
16:56:27 all 1,00 0,00 0,60 0,00 0,00 98,39

16:56:26 kbmemfree kbavail kbmemused %memused kbbuffers kbcached k
16:56:27 2428908 8846896 4554516 29,34 227228 7594048 239551

16:56:26 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:27 4194300 0 0,00 0 0,00

16:56:27 CPU %user %nice %system %iowait %steal %idle
16:56:28 all 1,61 0,00 0,65 0,05 0,00 97,68

```

```

16:56:27 kbmemfree kbavail kbmemused %memused kbbuffers kbcached k
16:56:28 2448788 8866784 4555236 29,34 227236 7573440 23935

16:56:27 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:28 4194300 0 0,00 0 0,00

16:56:28 CPU %user %nice %system %iowait %steal %idle
16:56:29 all 1,51 0,00 0,81 0,00 0,00 97,68

16:56:28 kbmemfree kbavail kbmemused %memused kbbuffers kbcached k
16:56:29 2456532 8874528 4564412 29,40 227236 7556520 239180

16:56:28 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:29 4194300 0 0,00 0 0,00

16:56:29 CPU %user %nice %system %iowait %steal %idle
16:56:30 all 1,71 0,00 0,76 0,10 0,00 97,43

16:56:29 kbmemfree kbavail kbmemused %memused kbbuffers kbcached k
16:56:30 2448432 8866436 4563784 29,40 227244 7565240 23927

16:56:29 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:30 4194300 0 0,00 0 0,00

16:56:30 CPU %user %nice %system %iowait %steal %idle
16:56:31 all 1,66 0,00 0,76 0,00 0,00 97,58

16:56:30 kbmemfree kbavail kbmemused %memused kbbuffers kbcached k
16:56:31 2449212 8867216 4562656 29,39 227244 7565588 239279

16:56:30 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:31 4194300 0 0,00 0 0,00

16:56:31 CPU %user %nice %system %iowait %steal %idle
16:56:32 all 1,62 0,00 0,91 0,15 0,00 97,32

```

```

16:56:31 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:32 2434132 8852152 4558212 29,36 227260 7585096 239475

16:56:31 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:32 4194300 0 0,00 0 0,00

16:56:32 CPU %user %nice %system %iowait %steal %idle
16:56:33 all 1,82 0,00 0,81 0,00 0,00 97,37

16:56:32 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:33 2428072 8846092 4563880 29,40 227260 7585488 23948

16:56:32 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:33 4194300 0 0,00 0 0,00

16:56:33 CPU %user %nice %system %iowait %steal %idle
16:56:34 all 5,89 0,00 0,81 0,00 0,00 93,30

16:56:33 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:34 2426444 8844520 4567940 29,42 227260 7583072 23943

16:56:33 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:34 4194300 0 0,00 0 0,00

16:56:34 CPU %user %nice %system %iowait %steal %idle
16:56:35 all 1,11 0,00 0,55 0,00 0,00 98,34

16:56:34 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:35 2400348 8818424 4580228 29,50 227260 7596880 23968

16:56:34 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:35 4194300 0 0,00 0 0,00

16:56:35 CPU %user %nice %system %iowait %steal %idle
16:56:36 all 1,20 0,00 0,75 0,00 0,00 98,04

```

```

16:56:35 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:36 2397868 8815944 4596296 29,61 227260 7583292 239550

16:56:35 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:36 4194300 0 0,00 0 0,00

16:56:36 CPU %user %nice %system %iowait %steal %idle
16:56:37 all 1,56 0,00 0,91 0,10 0,00 97,43

16:56:36 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:37 2391676 8809760 4593616 29,59 227268 7592164 2396418

16:56:36 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:37 4194300 0 0,00 0 0,00

16:56:37 CPU %user %nice %system %iowait %steal %idle
16:56:38 all 1,51 0,00 0,70 0,00 0,00 97,78

16:56:37 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:38 2414088 8832172 4586524 29,54 227268 7576844 239489

16:56:37 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:38 4194300 0 0,00 0 0,00

16:56:38 CPU %user %nice %system %iowait %steal %idle
16:56:39 all 1,16 0,00 0,86 0,00 0,00 97,98

16:56:38 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:39 2431924 8850008 4577696 29,49 227268 7567836 239400

16:56:38 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:39 4194300 0 0,00 0 0,00

16:56:39 CPU %user %nice %system %iowait %steal %idle
16:56:40 all 1,46 0,00 0,60 0,15 0,00 97,79

```

```

16:56:39 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:40 2430808 8848908 4570124 29,44 227284 7576508 23948

16:56:39 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:40 4194300 0 0,00 0 0,00

16:56:40 CPU %user %nice %system %iowait %steal %idle
16:56:41 all 1,31 0,00 0,81 0,00 0,00 97,88

16:56:40 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:41 2427880 8845976 4575184 29,47 227284 7574376 239468

16:56:40 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:41 4194300 0 0,00 0 0,00

16:56:41 CPU %user %nice %system %iowait %steal %idle
16:56:42 all 1,31 0,00 0,71 0,15 0,00 97,83

16:56:41 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:42 2419396 8837764 4582500 29,52 227300 7575552 239602

16:56:41 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:42 4194300 0 0,00 0 0,00

16:56:42 CPU %user %nice %system %iowait %steal %idle
16:56:43 all 2,12 0,00 0,91 0,15 0,00 96,82

16:56:42 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:43 2415508 8833148 4597812 29,62 227336 7564140 2393728

16:56:42 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:43 4194300 0 0,00 0 0,00

16:56:43 CPU %user %nice %system %iowait %steal %idle
16:56:44 all 1,56 0,00 0,76 0,15 0,00 97,53

```

```

16:56:43 kbmempfree kbavail kbmempused %mempused kbbuffers kbcached k
16:56:44 2417172 8834820 4600176 29,63 227352 7560096 239333

16:56:43 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:44 4194300 0 0,00 0 0,00

16:56:44 CPU %user %nice %system %iowait %steal %idle
16:56:45 all 1,36 0,00 0,75 0,00 0,00 97,89

16:56:44 kbmempfree kbavail kbmempused %mempused kbbuffers kbcached k
16:56:45 2415056 8832704 4598932 29,62 227352 7563460 23936!

16:56:44 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:45 4194300 0 0,00 0 0,00

16:56:45 CPU %user %nice %system %iowait %steal %idle
16:56:46 all 1,06 0,00 0,96 0,00 0,00 97,99

16:56:45 kbmempfree kbavail kbmempused %mempused kbbuffers kbcached k
16:56:46 2420972 8838628 4592440 29,58 227352 7564036 23937

16:56:45 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:46 4194300 0 0,00 0 0,00

16:56:46 CPU %user %nice %system %iowait %steal %idle
16:56:47 all 1,36 0,00 0,40 0,05 0,00 98,18

16:56:46 kbmempfree kbavail kbmempused %mempused kbbuffers kbcached k
16:56:47 2427264 8844920 4584996 29,53 227352 7565188 23938.

16:56:46 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:47 4194300 0 0,00 0 0,00

16:56:47 CPU %user %nice %system %iowait %steal %idle
16:56:48 all 1,10 0,00 0,70 0,00 0,00 98,19

```

```

16:56:47 kbmempfree kbavail kbmempused %mempused kbbuffers kbcached k
16:56:48 2432900 8850556 4583096 29,52 227352 7561452 23934

16:56:47 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:48 4194300 0 0,00 0 0,00

16:56:48 CPU %user %nice %system %iowait %steal %idle
16:56:49 all 0,86 0,00 0,45 0,05 0,00 98,64

16:56:48 kbmempfree kbavail kbmempused %mempused kbbuffers kbcached k
16:56:49 2427216 8844872 4579252 29,50 227360 7570972 239442

16:56:48 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:49 4194300 0 0,00 0 0,00

16:56:49 CPU %user %nice %system %iowait %steal %idle
16:56:50 all 1,21 0,00 0,66 0,00 0,00 98,13

16:56:49 kbmempfree kbavail kbmempused %mempused kbbuffers kbcached k
16:56:50 2424084 8841752 4581384 29,51 227360 7571972 2394528

16:56:49 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:50 4194300 0 0,00 0 0,00

16:56:50 CPU %user %nice %system %iowait %steal %idle
16:56:51 all 1,56 0,00 0,91 0,00 0,00 97,53

16:56:50 kbmempfree kbavail kbmempused %mempused kbbuffers kbcached k
16:56:51 2431264 8848936 4577068 29,48 227360 7569108 239423

16:56:50 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:51 4194300 0 0,00 0 0,00

16:56:51 CPU %user %nice %system %iowait %steal %idle
16:56:52 all 3,09 0,00 1,37 0,00 0,00 95,54

```

```

16:56:51 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:52 2286456 8704128 4579552 29,50 227360 7711400 240855

16:56:51 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:52 4194300 0 0,00 0 0,00

16:56:52 CPU %user %nice %system %iowait %steal %idle
16:56:53 all 1,11 0,00 0,81 0,05 0,00 98,03

16:56:52 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:53 2418636 8836308 4584600 29,53 227360 7574172 239482

16:56:52 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:53 4194300 0 0,00 0 0,00

16:56:53 CPU %user %nice %system %iowait %steal %idle
16:56:54 all 1,41 0,00 0,86 0,00 0,00 97,73

16:56:53 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:54 2442144 8859816 4585140 29,53 227360 7550140 239242

16:56:53 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:54 4194300 0 0,00 0 0,00

16:56:54 CPU %user %nice %system %iowait %steal %idle
16:56:55 all 1,36 0,00 0,90 0,10 0,00 97,64

16:56:54 kbmempfree kbavail kbmempused %memused kbbuffers kbcached k
16:56:55 2483312 8900992 4566656 29,42 227368 7527448 23890.

16:56:54 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:55 4194300 0 0,00 0 0,00

16:56:55 CPU %user %nice %system %iowait %steal %idle
16:56:56 all 5,36 0,00 0,91 0,05 0,00 93,67

```

```

16:56:55 kbmemfree kbavail kbmemused %memused kbbuffers kbcached k
16:56:56 2563616 8981296 4575008 29,47 227368 7438792 238026

16:56:55 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:56 4194300 0 0,00 0 0,00

16:56:56 CPU %user %nice %system %iowait %steal %idle
16:56:57 all 1,26 0,00 0,65 0,00 0,00 98,09

16:56:56 kbmemfree kbavail kbmemused %memused kbbuffers kbcached k
16:56:57 2565368 8983048 4570892 29,44 227368 7441156 238049

16:56:56 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:57 4194300 0 0,00 0 0,00

16:56:57 CPU %user %nice %system %iowait %steal %idle
16:56:58 all 0,96 0,00 0,60 0,00 0,00 98,44

16:56:57 kbmemfree kbavail kbmemused %memused kbbuffers kbcached k
16:56:58 2602268 9019948 4569656 29,44 227368 7405492 23769

16:56:57 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:58 4194300 0 0,00 0 0,00

16:56:58 CPU %user %nice %system %iowait %steal %idle
16:56:59 all 1,21 0,00 0,50 0,05 0,00 98,24

16:56:58 kbmemfree kbavail kbmemused %memused kbbuffers kbcached k
16:56:59 2597520 9015200 4571152 29,44 227368 7408744 237726

16:56:58 kbswpfree kbswpused %swpused kbswpcad %swpcad
16:56:59 4194300 0 0,00 0 0,00

Media: CPU %user %nice %system %iowait %steal %idle
Media: all 1,40 0,00 0,68 0,04 0,00 97,88

```

```
Media:  kbmempfree  kbavail kbmempused  %memused kbbuffers kbcached kl
Media:    2454669  8872565  4572474    29,45  227258  7550340  239196
```

```
Media:  kbswpfree kbswpused  %swpused kbswpcad  %swpcad
Media:    4194300      0    0,00      0    0,00
```

3.3 Carga real

Procedemos a instalar en el sistema `ffmpeg`:

```
ucian@lucian-Blade-15-2022-RZ09-0421:~$ sudo apt update
sudo apt install ffmpeg
Obj:1 http://archive.ubuntu.com/ubuntu oracular InRelease
Obj:2 http://security.ubuntu.com/ubuntu oracular-security InRelease
Obj:3 https://brave-browser-apt-release.s3.brave.com stable InRelease
Obj:4 http://archive.ubuntu.com/ubuntu oracular-updates InRelease
Obj:5 http://archive.ubuntu.com/ubuntu oracular-backports InRelease
Se pueden actualizar 88 paquetes. Ejecute «apt list --upgradable» para verlos.
Installing:
  ffmpeg
```

Installing dependencies:

```
i965-va-driver      libhwy1t64         libsratom-0-0
intel-media-va-driver libigdgmm12        libsrt1.5-gnutls
libaacs0            libjxl0.10         libsvtav1enc2
libass9             liblapack3         libswresample5
libavcodec61        liblilv-0-0        libswscale8
libavdevice61       libmbedcrypto7t64  libudfread0
libavfilter10       libmysofa1         libunibreak6
libavformat61       libnorm1t64        libva-drm2
libavutil59         libopenal-data     libva-x11-2
libbdplus0          libopenal1         libva2
libblas3            libopenmpt0t64     libvidstab1.1
```

```
libbluray2      libpgm-5.3-0t64  libvpl2
libbs2b0        libplacebo349    libx264-164
libchromaprint1 libpocketsphinx3 libx265-209
libcjson1       libpostproc58    libxvidcore4
libcodec2-1.2   librabbitmq4     libzimg2
libdav1d7       librav1e0.7      libzix-0-0
libdc1394-25    librist4         libzmq5
libdvdnav4      librubberband2   libzvbi-common
libdvdread8t64  libserd-0-0      libzvbi0t64
libfftw3-double3 libshine3        mesa-va-drivers
libflite1       libsndio7.0      ocl-icd-libopenc1
libgfortran5    libsord-0-0      pocketsphinx-en-us
libgme0         libsoxr0         va-driver-all
libgsm1         libsphinxbase3t64
```

Paquetes sugeridos:

```
ffmpeg-doc      libnvidia-encode1 libfftw3-bin  serdi
i965-va-driver-shaders libbluray-bdj    libfftw3-dev  sndiod
libnvcuvid1     libdvdcss2      libportaudio2 sordi
```

Summary:

```
Upgrading: 0, Installing: 75, Removing: 0, Not Upgrading: 88
Download size: 89,9 MB
Space needed: 217 MB / 99,7 GB available
```

Continue? [S/n] S

Usaremos el enlace proporcionado para la descarga del video:

```
lucian@lucian-Blade-15-2022-RZ09-0421:~$ wget http://distribution.bbb3d.renderfarming.net/video/mp4
--2025-03-18 17:06:07-- http://distribution.bbb3d.renderfarming.net/video/mp4
Resolviendo distribution.bbb3d.renderfarming.net (distribution.bbb3d.renderfarming.net)
Conectando con distribution.bbb3d.renderfarming.net (distribution.bbb3d.renderfarming.net)
Petición HTTP enviada, esperando respuesta... 200 OK
Longitud: 276134947 (263M) [video/mp4]
Guardando como: 'bbb_sunflower_1080p_30fps_normal.mp4'
```

bbb_sunflower_1080p 100%[=====⇒] 263,34M 5,92MB/s en

2025-03-18 17:07:35 (3,02 MB/s) - 'bbb_sunflower_1080p_30fps_normal.mp4' gu

Ejecutaremos el `ffmpeg` para generar la carga usando este script en bash:

```
#!/bin/bash

# Ejecutamos ffmpeg
ffmpeg -y -i bbb_sunflower_1080p_30fps_normal.mp4 -preset fast bbb_transcod
PID=$!

SECONDS=0
echo "[" > usage_ffmpeg.json

first_entry=true

# Monitorización cada segundo
while kill -0 $PID 2>/dev/null; do
    CPU=$(ps -p $PID -o %cpu= | tr -d ' ')
    MEM=$(ps -p $PID -o %mem --no-headers | tr -d ' ')

    # Validar que CPU y MEM no estén vacíos
    if [ -z "$CPU" ]; then CPU="0.0"; fi
    if [ -z "$MEM" ]; then MEM="0.0"; fi

    # Añadir coma solo después del primer registro
    if [ "$SECONDS" -ne 0 ]; then
        echo "," >> usage_ffmpeg.json
    fi

    # Guardar datos en formato JSON
    echo "{ \"time\": $SECONDS, \"cpu\": $CPU, \"memory\": $MEM }" >> usage_f

    sleep 1
```

```
SECONDS=$((SECONDS + 1))
done
```

```
echo "Monitorización finalizada. Resultados guardados en usage_ffmpeg.json"
```

Lo guardamos en un `.json` por preferencia personal para obtener las tablas y verlas en jupyter notebook.

```
import pandas as pd
import matplotlib.pyplot as plt
```

1) DATOS INICIALES del primer comando

```
initial_data = {
    "time": list(range(60)),
    "cpu_idle": [
        98.74, 98.39, 98.49, 98.14, 98.30, 98.64, 98.09, 98.54, 98.30, 98.34,
        98.59, 98.29, 98.40, 98.64, 98.09, 98.54, 98.30, 98.34, 98.59, 98.29,
        98.40, 98.64, 98.09, 98.54, 98.30, 98.34, 98.59, 98.29, 98.40, 98.64,
        98.09, 98.54, 98.30, 98.34, 93.30, 98.29, 98.03, 98.29, 98.59, 98.14,
        97.68, 97.68, 97.43, 97.58, 97.32, 97.37, 97.53, 97.89, 97.99, 98.18,
        98.03, 97.73, 97.64, 93.67, 98.09, 98.44, 98.24, 97.88, 97.78, 98.03,
    ],
    "mem_used": [
        29.38, 29.39, 29.38, 29.37, 29.36, 29.36, 29.43, 29.45, 29.46, 29.42,
        29.41, 29.42, 29.43, 29.43, 29.44, 29.44, 29.45, 29.46, 29.46, 29.44,
        29.41, 29.42, 29.40, 29.39, 29.38, 29.37, 29.36, 29.43, 29.45, 29.46,
        29.42, 29.41, 29.42, 29.43, 29.43, 29.44, 29.44, 29.45, 29.46, 29.46,
        29.44, 29.41, 29.42, 29.40, 29.39, 29.49, 29.31, 29.35, 29.34, 29.34,
        29.40, 29.40, 29.39, 29.40, 29.39, 29.36, 29.50, 29.42, 29.61, 29.59
    ]
}
```

```
initial_df = pd.DataFrame(initial_data)
initial_df["cpu_usage"] = 100 - initial_df["cpu_idle"]
```

```
initial_df.rename(columns={"mem_used": "mem_usage"}, inplace=True) # Memc
```

2) DATOS FFmpeg

```
data_ffmpeg = [  
    {"time": 0, "cpu": 0.0, "memory": 0.0 },  
    {"time": 2, "cpu": 939, "memory": 8.0 },  
    {"time": 4, "cpu": 1272, "memory": 8.0 },  
    {"time": 6, "cpu": 1406, "memory": 8.0 },  
    {"time": 8, "cpu": 1471, "memory": 8.0 },  
    {"time": 10, "cpu": 1472, "memory": 8.0 },  
    {"time": 12, "cpu": 1513, "memory": 8.0 },  
    {"time": 15, "cpu": 1544, "memory": 8.0 },  
    {"time": 17, "cpu": 1563, "memory": 8.0 },  
    {"time": 19, "cpu": 1579, "memory": 8.0 },  
    {"time": 21, "cpu": 1584, "memory": 8.0 },  
    {"time": 23, "cpu": 1590, "memory": 8.0 },  
    {"time": 25, "cpu": 1591, "memory": 8.0 },  
    {"time": 27, "cpu": 1579, "memory": 8.0 },  
    {"time": 29, "cpu": 1576, "memory": 8.0 },  
    {"time": 31, "cpu": 1567, "memory": 8.0 },  
    {"time": 33, "cpu": 1571, "memory": 8.0 },  
    {"time": 35, "cpu": 1581, "memory": 8.0 },  
    {"time": 37, "cpu": 1579, "memory": 8.0 },  
    {"time": 39, "cpu": 1576, "memory": 8.0 },  
    {"time": 41, "cpu": 1564, "memory": 8.0 },  
    {"time": 43, "cpu": 1554, "memory": 8.0 },  
    {"time": 45, "cpu": 1551, "memory": 8.0 },  
    {"time": 47, "cpu": 1555, "memory": 8.0 },  
    {"time": 49, "cpu": 1555, "memory": 8.1 },  
    {"time": 51, "cpu": 1555, "memory": 8.1 },  
    {"time": 53, "cpu": 1556, "memory": 8.1 },  
    {"time": 55, "cpu": 1547, "memory": 8.1 },  
    {"time": 57, "cpu": 1547, "memory": 8.1 },  
    {"time": 59, "cpu": 1546, "memory": 8.1 },
```

```
{ "time": 61, "cpu": 1543, "memory": 8.1 },  
{ "time": 63, "cpu": 1542, "memory": 8.1 },  
{ "time": 65, "cpu": 1541, "memory": 8.1 },  
{ "time": 67, "cpu": 1540, "memory": 8.1 },  
{ "time": 70, "cpu": 1544, "memory": 8.1 },  
{ "time": 72, "cpu": 1545, "memory": 8.1 },  
{ "time": 74, "cpu": 1542, "memory": 8.1 },  
{ "time": 76, "cpu": 1541, "memory": 8.1 },  
{ "time": 78, "cpu": 1541, "memory": 8.1 },  
{ "time": 80, "cpu": 1536, "memory": 8.1 },  
{ "time": 82, "cpu": 1535, "memory": 8.1 },  
{ "time": 84, "cpu": 1529, "memory": 8.1 },  
{ "time": 86, "cpu": 1531, "memory": 8.1 },  
{ "time": 88, "cpu": 1529, "memory": 8.1 },  
{ "time": 90, "cpu": 1533, "memory": 8.1 },  
{ "time": 92, "cpu": 1533, "memory": 8.1 },  
{ "time": 94, "cpu": 1533, "memory": 8.1 },  
{ "time": 96, "cpu": 1532, "memory": 8.1 },  
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```

```
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{ "time": 205, "cpu": 1529, "memory": 8.3 },  
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```

```
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{ "time": 266, "cpu": 1527, "memory": 8.4 },  
{ "time": 268, "cpu": 1526, "memory": 8.4 }  
]
```

```
ffmpeg_df = pd.DataFrame(data_ffmpeg)
```

```
ffmpeg_df["cpu"] = ffmpeg_df["cpu"] / 100
```

GRÁFICA 1: COMPARATIVA CPU

```
plt.figure(figsize=(12, 6))
plt.plot(initial_df["time"], initial_df["cpu_usage"], label="CPU Idle ⇒ Uso (Estado
plt.plot(ffmpeg_df["time"], ffmpeg_df["cpu"], label="Uso CPU (FFmpeg)", marker=
plt.xlabel("Tiempo (s)")
plt.ylabel("Uso de CPU (%)")
plt.title("Comparativa de Uso de CPU (Idle vs. FFmpeg)")
plt.legend()
plt.grid()
plt.show()
```

GRÁFICA 2: COMPARATIVA MEMORIA

```
idle_mem_mean = initial_df["mem_usage"].mean()
ffmpeg_df["memory_shifted"] = ffmpeg_df["memory"] + idle_mem_mean

plt.figure(figsize=(12, 6))
plt.plot(
    initial_df["time"],
    initial_df["mem_usage"],
    label="Memoria (Estado Inicial)",
    marker="o"
)
plt.plot(
    ffmpeg_df["time"],
    ffmpeg_df["memory_shifted"],
    label=f"Memoria (FFmpeg) + Idle Mean ({idle_mem_mean:.2f}%)",
    marker="o",
    color="orange"
)
plt.xlabel("Tiempo (s)")
plt.ylabel("Uso de Memoria (%)")
```

```

plt.title("Comparativa de Uso de Memoria (Idle vs. FFmpeg+Mean)")
plt.legend()
plt.grid()
plt.show()

# TABLA RESUMEN CPU

summary_cpu = pd.DataFrame({
    "Medición": ["Estado Inicial", "Durante FFmpeg"],
    "CPU Promedio (%)": [initial_df["cpu_usage"].mean(), ffmpeg_df["cpu"].mean()],
    "CPU Máximo (%)": [initial_df["cpu_usage"].max(), ffmpeg_df["cpu"].max()],
    "CPU Mínimo (%)": [initial_df["cpu_usage"].min(), ffmpeg_df["cpu"].min()],
})

print("\nTabla Resumen del Uso de CPU:")
print(summary_cpu)

# TABLA RESUMEN MEMORIA (IDLE)

summary_mem_init = pd.DataFrame({
    "Uso Memoria Promedio (%)": [initial_df["mem_usage"].mean()],
    "Uso Memoria Máximo (%)": [initial_df["mem_usage"].max()],
})

print("\nResumen de Memoria (Estado Inicial):")
print(summary_mem_init)

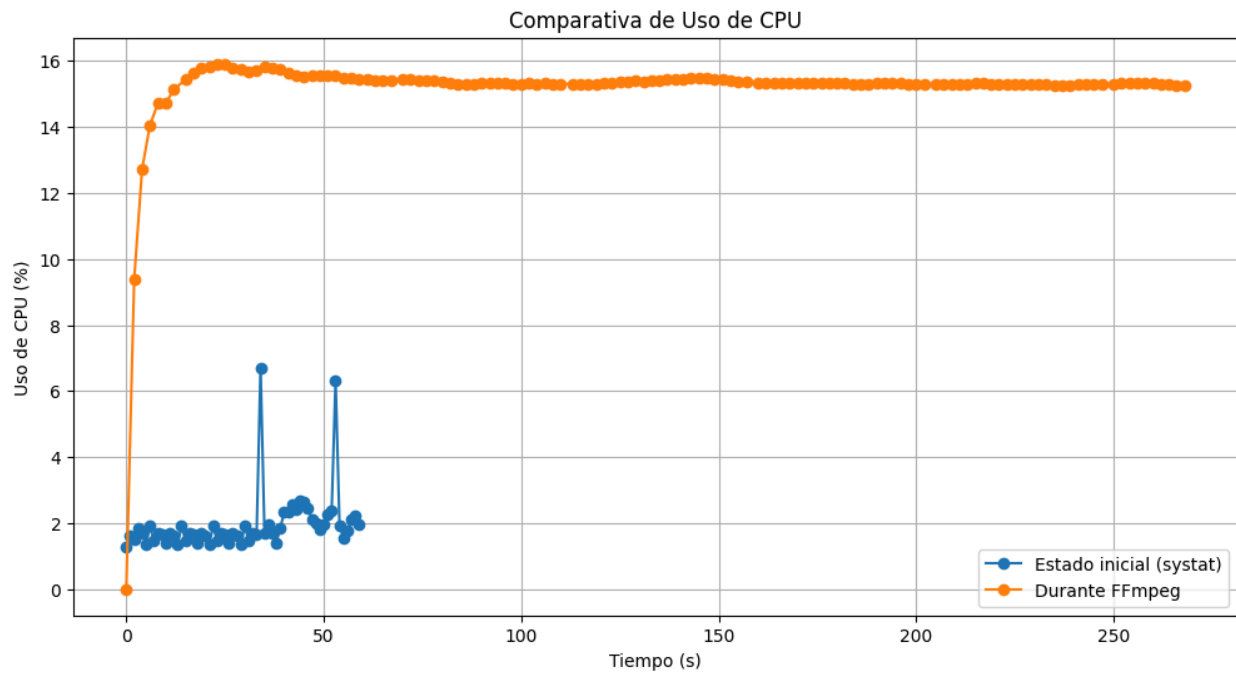
# TABLA RESUMEN MEMORIA (FFmpeg)

summary_mem_ffmpeg_shifted = pd.DataFrame({
    "Uso Memoria Promedio (IdleMean + FFmpeg)": [ffmpeg_df["memory_shifted"].mean()],
    "Uso Memoria Máximo (IdleMean + FFmpeg)": [ffmpeg_df["memory_shifted"].max()],
})

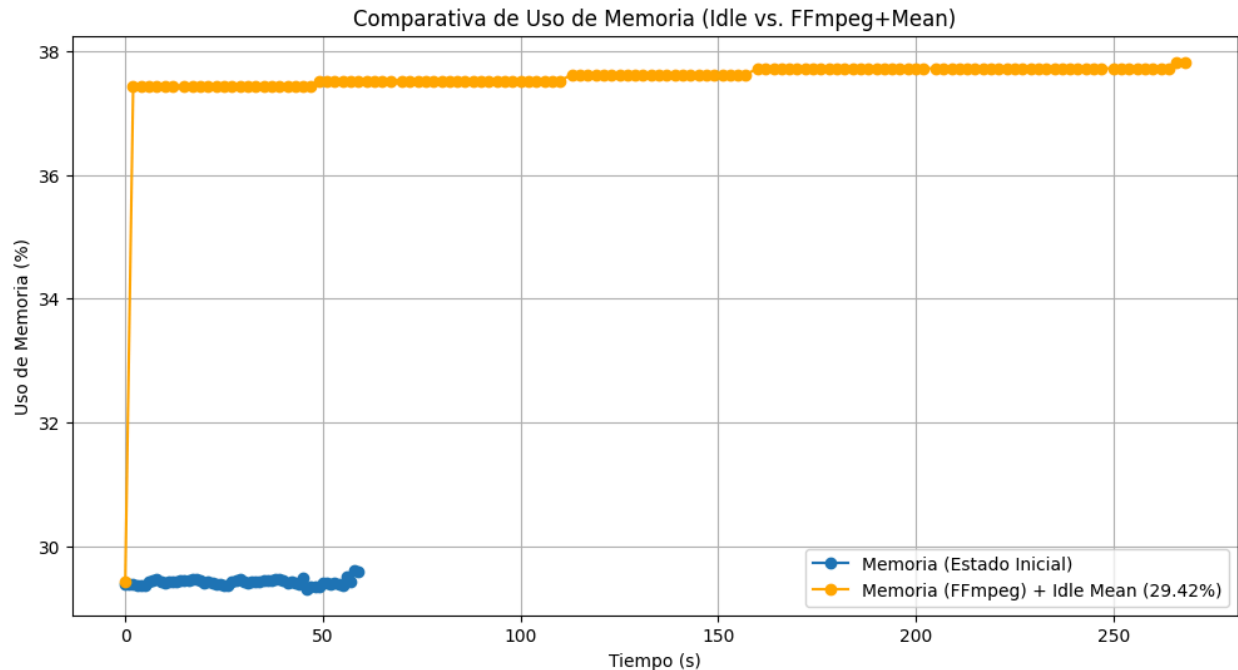
```

```
})
```

```
print("\nResumen de Memoria (FFmpeg + IdleMean):")  
print(summary_mem_ffmpeg_shifted)
```



Comparativa uso de CPU de prueba inicial frente a la ejecución Ffmpeg



Comparativa uso de Memoria del estado inicial frente al uso de Ffmpeg

4. Métricas Seleccionadas

- Carga CPU (% de uso).
- Consumo de Memoria RAM (MB usados, % de uso).

5. Herramientas Utilizadas

Para monitorizar y analizar el rendimiento del sistema, se han empleado las siguientes herramientas:

1. **sysstat** (comando `sar`): Permite registrar a intervalos regulares el uso de CPU, memoria y otros parámetros del sistema. Hemos usado: para monitorizar CPU (`u`), memoria (`r`), y paginación/swap (`s`) durante 1 segundo de intervalo con 60 muestras.

```
sar -u -r -S 1 60
```

1. **free** y **/proc/meminfo**: Para comprobar de manera directa la memoria disponible y su uso (%memused), así como para corroborar los datos que presenta `sar`.

2. **df y uname:** Para información de almacenamiento (`df -h`) y versión del kernel (`uname -r`) junto con información de la máquina.
3. **ffmpeg** (carga real): Utilizado para generar un proceso intensivo de CPU y algo de E/S de disco (transcodificación de vídeo). El script en bash se encarga de lanzar `ffmpeg` y capturar su PID para monitorizar el %CPU y %MEM en formato JSON cada segundo.
4. **Python (pandas, matplotlib):** Procesado de datos y visualización de los resultados (gráficos de uso de CPU y memoria), mediante Jupyter Notebook.

Con estas herramientas combinadas, hemos obtenido una **radiografía** del estado del sistema en un escenario sin carga y bajo carga, permitiendo compararlas.

6. Conclusiones

A partir de la monitorización y el análisis realizados, se pueden destacar los siguientes puntos:

1. Estado Ocioso vs. Bajo Carga

En estado ocioso, la CPU se mantiene con altos valores de *idle* (en torno al 97–99%), lo que indica un bajo consumo de recursos cuando el sistema no ejecuta tareas pesadas. Al lanzar la transcodificación de vídeo con `ffmpeg`, se observa un incremento significativo en el uso de CPU (llegando a valores cercanos a 15–16% en la escala elegida) y un incremento moderado en la memoria (alrededor de un 8% sobre el total).

2. Uso de Memoria

Durante el estado inicial, la memoria usada se sitúa en torno al 29–30%. Al ejecutar FFmpeg, si desplazamos su uso un valor equivalente a la media del estado ocioso, vemos que la "memoria total" para el proceso + idle estaría en torno a 37–38%.

Esto indica que, aunque el proceso de transcodificación aumenta el uso de CPU, el incremento de memoria no es tan agresivo.

3. Herramientas Efectivas

`sar -u -r -S` en modo batch resultó útil para registrar las estadísticas de CPU y memoria de forma continua a lo largo de 1 minuto. El script en bash para

capturar los datos de `ffmpeg` (PID) y exportarlos a JSON simplificó el análisis posterior en Python.

4. Posibles Cuellos de Botella

En la transcodificación, la CPU se ve más exigida que la memoria. El disco no pareció ser un cuello de botella notable en este experimento, aunque con mayor volumen o distintos códecs podría aumentar la presión en I/O.

5. Recomendaciones que se nos ocurren

En sistemas con recursos similares, la transcodificación de vídeo no impacta tanto en memoria, pero puede elevar el uso de CPU de forma significativa. Para cargas más largas, podría ser interesante evaluar la temperatura del sistema y su capacidad de disipación para evitar los “bloqueos” del sistema en procesos de larga duración.